# Mixed Linear Models for Association Genetics .

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# Models for GWAS

## Data sets

$$\underset{(n,1)}{\begin{matrix} Y \\ \begin{pmatrix} 15 \\ 22 \\ \vdots \\ 33 \end{pmatrix} \end{matrix}} = \underset{(n,845637)}{\begin{matrix} SNP_1 & SNP_2 & \dots & SNP_{845637} \\ \begin{pmatrix} A & C & \dots & C \\ G & T & \dots & C \\ \vdots & \vdots & & \vdots \\ A & T & \dots & T \end{pmatrix} \end{matrix}} \quad \text{et} \quad \underset{(n,1)}{\begin{matrix} Pops \\ \begin{pmatrix} P1 \\ P1 \\ \vdots \\ P8 \end{pmatrix} \end{matrix}}$$

# Matrix algebra

$\Longrightarrow$ for **each** SNP $\Longleftarrow$

$$\underset{(n,1)}{\begin{pmatrix} 15 \\ 22 \\ \vdots \\ 33 \end{pmatrix}} = \underset{(n,2)}{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 1 & 0 \end{pmatrix}} \cdot \underset{(2,1)}{\begin{pmatrix} \mathrm{A} \\ \mathrm{G} \end{pmatrix}}$$

$$\begin{matrix} Y \\ \begin{pmatrix} 15 \\ 22 \\ \vdots \\ 33 \end{pmatrix} \\ (n,1) \end{matrix} = \begin{matrix} \mu & a_1 & a_2 \\ \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \\ 1 & 1 & 0 \end{pmatrix} \\ (n,3) \end{matrix} \cdot \underset{(2,1)}{\begin{pmatrix} \mu \\ a_1 \\ a_2 \end{pmatrix}} + \underset{(n,1)}{\mathbf{e}}$$

# Matrix algebra

$\Longrightarrow$ for **each** SNP $\Longleftarrow$

$$\underset{(n,1)}{\begin{matrix} Y \\ \begin{pmatrix} 15 \\ 22 \\ \vdots \\ 33 \end{pmatrix} \end{matrix}} = \underset{(n,2)}{\begin{matrix} \mu & a_1 \\ \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ \vdots & \vdots \\ 1 & 1 \end{pmatrix} \end{matrix}} \cdot \underset{(2,1)}{\begin{pmatrix} \mu \\ a_1 \end{pmatrix}} + \underset{(n,1)}{\mathbf{e}} \qquad \text{and} \qquad a_2 = -a_1$$

$$\underset{(n,1)}{\mathbf{Y}} = \underset{(n,2)}{\mathbf{X}} \cdot \underset{(2,1)}{\begin{pmatrix} \mu \\ a_1 \end{pmatrix}} + \underset{(n,1)}{\mathbf{e}}$$

# Fixed model for GWAS

The basic model is a linear fixed model with effects of genotype at SNP :

$$\underset{(n,1)}{\mathbf{Y}} = \underset{(n,2)}{\mathbf{X}} \cdot \underset{(2,1)}{\begin{pmatrix} \mu \\ a \end{pmatrix}} + \underset{(n,1)}{\mathbf{e}}$$
$$= \underset{(n,2)}{\mathbf{X}} \cdot \underset{(2,1)}{\mathbf{b}} + \underset{(n,1)}{\mathbf{e}}$$

- If 845637 SNPs are assessed, 845637 models are successively fitted !

SNPs are numerous, thus the recombination fraction between a SNP and a QTL located near the SNP is low.
Estimations of **a** and **d** are thus little biaised.

# Matrix algebra

$\Longrightarrow$ for **each** SNP $\Longleftarrow$, a 'population' **fixed** factor may be added

$$\underset{(n,1)}{\begin{pmatrix} 15 \\ 22 \\ \vdots \\ 33 \end{pmatrix}} = \underset{(n,2+p-1)}{\begin{pmatrix} \mu & a_1 & b_1 & b_2 & \dots & b_{p-1} \\ 1 & 1 & 1 & 0 & \dots & 0 \\ 1 & -1 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & & & & \\ 1 & 1 & -1 & -1 & \dots & -1 \end{pmatrix}} \cdot \underset{2+p-1,1}{\begin{pmatrix} \mu \\ a_1 \\ b_1 \\ \vdots \\ b_{p-1} \end{pmatrix}} + \underset{(n,1)}{\mathbf{e}}$$

and

$$\begin{cases} a_2 &= -a_1 \\ b_p &= -\sum b_j \end{cases}$$

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It is also possible to fit the components of the experimental design (blocks, sites, years,...) as fixed effects.

# Mixed Linear Model for **each** SNP : relatedness and unbalanced designs.

- If $g$ genotypes, from $p$ populations, are assessed one time :
  - SNP effects $(\mu, a_1)$ **fixed**
  - population effects $(b_1 \dots b_{p-1})$ **fixed**
  - general genetic effect of each accession $(u_1 \dots u_g)$ **random** – "random deviates due to genome-wide relatedness"

$$\underset{(g,1)}{\mathbf{Y}} = \underset{(g,2+p-1)}{\mathbf{X}} \cdot \underset{(2+p-1,1)}{\begin{pmatrix} \mu \\ a_1 \\ b_1 \\ \vdots \\ b_{p-1} \end{pmatrix}} + \underset{(g,g)}{\mathbf{Z}} \cdot \underset{(g,1)}{\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_g \end{pmatrix}} + \underset{(g,1)}{\mathbf{e}}$$

$$\underset{(g,1)}{\mathbf{Y}} = \underset{(g,2+p-1)}{\mathbf{X}} \cdot \underset{(2+p-1,1)}{\mathbf{b}} + \underset{(g,g)}{\mathbf{Z}} \cdot \underset{(g,1)}{\mathbf{u}} + \underset{(g,1)}{\mathbf{e}}$$

# A mixed linear model is fitted for **each** SNP, to account for relatedness among accessions

- if $g$ genotypes, in $p$ populations, are assessed within $q$ environnements ($n = g \times q$ measurements) :
  - SNP effects $(\mu, a_1)$ **fixed**
  - Population effects $(b_1 \dots b_{p-1})$ **fixed** and environment effects $(c_1 \dots c_{q-1})$ **fixed**
  - Genetic of each accession $(u_1 \dots u_g)$ **random**

$$\underset{(n,1)}{\mathbf{Y}} = \underset{(n,2+(p-1)+(q-1))}{\mathbf{X}} \cdot \underset{(2+(p-1)+(q-1),1)}{\begin{pmatrix} \mu \\ a_1 \\ b_1 \\ \vdots \\ b_{p-1} \\ c_1 \\ \vdots \\ c_{q-1} \end{pmatrix}} + \underset{(n,g)}{\mathbf{Z}} \cdot \underset{(g,1)}{\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_g \end{pmatrix}} + \underset{(n,1)}{\mathbf{e}}$$

# Henderson Equations

$$\begin{pmatrix} {}^tXR^{-1}X & {}^tXR^{-1}Z \\ {}^tZR^{-1}X & {}^tZR^{-1}Z + G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} {}^tXR^{-1}y \\ {}^tZR^{-1}y \end{pmatrix}$$

with :

$$\begin{bmatrix} \mathbf{u} \\ \mathbf{e} \end{bmatrix} = \mathcal{N}(\mathbf{0}, \begin{bmatrix} \mathbf{G} & \mathbf{0} \\ \mathbf{0} & \mathbf{R} \end{bmatrix})$$

We have to make hypotheses for **G** and **R**.

# Henderson Equations

simplifying hypotheses : $\mathbf{R} = \sigma^2.\mathbf{I}$ et $\mathbf{G} = \sigma^2_g.\mathbf{G}_{matrix}$

$$\begin{pmatrix} {}^tX.\frac{1}{\sigma^2_e}I.X & {}^tX.\frac{1}{\sigma^2_e}I.Z \\ {}^tZ.\frac{1}{\sigma^2_e}I.X & {}^tZ.\frac{1}{\sigma^2_e}I.Z + \frac{1}{\sigma^2_g}G^{-1}_{matrix} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} {}^tX.\frac{1}{\sigma^2_e}I.y \\ {}^tZ.\frac{1}{\sigma^2_e}I.y \end{pmatrix}$$

factor by $\frac{1}{\sigma^2_e}$ :

$$\frac{1}{\sigma^2_e}\begin{pmatrix} {}^tXX & {}^tXZ \\ {}^tZX & {}^tZZ + \frac{\sigma^2_e}{\sigma^2_g}G^{-1}_{matrix} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix} = \frac{1}{\sigma^2_e}\begin{pmatrix} {}^tXy \\ {}^tZy \end{pmatrix}$$

$$\begin{pmatrix} {}^tXX & {}^tXZ \\ {}^tZX & {}^tZZ + \lambda G^{-1}_{matrix} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} {}^tXy \\ {}^tZy \end{pmatrix}$$

with $\lambda = \frac{\sigma^2_e}{\sigma^2_g}$.

# Different types for $\mathbf{G}_{matrix}$, the variance-covariance matrix for genotypes

- nothing is known, or there is no relatedness among accessions :

$$\underset{(g,g)}{\mathbf{G}} = \sigma^2_g \cdot \underset{(g,g)}{\mathbf{I}}$$

- the pedigree matrix **A**, is known :

$$\underset{(g,g)}{\mathbf{G}} = \sigma^2_g \cdot \underset{(g,g)}{\mathbf{A}} \qquad \text{ex :} \quad \mathbf{A} = \begin{array}{c} \\ g_1 \\ g_2 \\ g_3 \end{array} \begin{array}{c} \begin{array}{ccc} g_1 & g_2 & g_3 \end{array} \\ \begin{pmatrix} 1 & 0.5 & 0.25 \\ 0.5 & 1 & 0.75 \\ 0.25 & 0.75 & 1 \end{pmatrix} \end{array}$$

# Different types for $\mathbf{G}_{matrix}$, the variance-covariance matrix for genotypes

- SNPs allow to compute **K**, the kinship matrix between accessions :

$$\underset{(g,g)}{\mathbf{G}} = \sigma^2_g \cdot \underset{(g,g)}{\mathbf{K}} \qquad \text{ex :} \quad \mathbf{K} = \begin{array}{c} \\ g_1 \\ g_2 \\ g_3 \end{array} \begin{array}{c} \begin{array}{ccc} g_1 & g_2 & g_3 \end{array} \\ \begin{pmatrix} 2 & 0.66 & 0.33 \\ 0.66 & 2 & 0.81 \\ 0.33 & 0.81 & 2 \end{pmatrix} \end{array}$$

One way to compute 'kinship coefficient' is :

$$f_{i,j} = \sum_k \frac{(x_{i,k} - p_k)(x_{j,k} - p_k)}{(p_k \times (1 - p_k))}$$

where $k$ changes from 1 to $N =$ the number of SNPs, $x_{i,k}$ is the genotype of the $i$th accession at the $k$th SNP, encoded as $0, 0.5, 1$, and $p_k$ the frequence of the reference allele. The diagonal is $0.5 \times (1 + inbreeding)$.

# Henderson's Equations

One additional simplification : $\mathbf{R} = \sigma^2.\mathbf{I}$ et $\mathbf{G} = \sigma^2_g.\mathbf{I}$ (case when using `lme4::lmer`)

$$\begin{pmatrix} {}^tXX & {}^tXZ \\ {}^tZX & {}^tZZ + \lambda G^{-1}_{matrix} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} {}^tXy \\ {}^tZy \end{pmatrix}$$

simplifies to :

$$\begin{pmatrix} {}^tXX & {}^tXZ \\ {}^tZX & {}^tZZ + \lambda I \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} {}^tXy \\ {}^tZy \end{pmatrix}$$

# Successive analyses using TASSEL

- Fixed Linear Model with only a SNP effect
- **Q** model : includes fixed effects of population structure.
- $\mathbf{K} + \mathbf{Q}$ model : MLM with fixed effects of SNP and population structure effet random effect for genotypes. The kinship matrix **K** is the variance-covariance matrix of accessions.

# Basis for TASSEL (and GAPIT)

- first implementation of the use of MLM for GWAS

  Yu, J. et al. (2006) Nature Genetics 38 :203–208.
- Optimisation of computation algorithms to solve Henderson equations – EMMA then EMMAX –

  Kang, H. M. et al. (2008) Genetics 178 :1709–1723 ;

  Kang, H. M. et al. (2010) Nature Genetics 42 :348–354.
- compressed MLM – CMLM – reduce the size of $\mathbf{K}$ by grouping accessions in clusters, using k-means and hierarchical clustering from SNPs data
- population parameters previously determined – P3D – use heritability computed at the phenotype level, to avoid (re) computation of $\lambda$ at each SNP.

  Zhang, Z. et al. (2010) Nature Genetics 42, 355–360.